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# Build or Adapt?
# Tests of Foundation Models on Survey Tiles

## The Question

Astronomical survey images arrive as tiles, with objects anywhere. Do we build a new tile-native model, or adapt a general one (DINOv3 plus a small adapter)?

## What we did

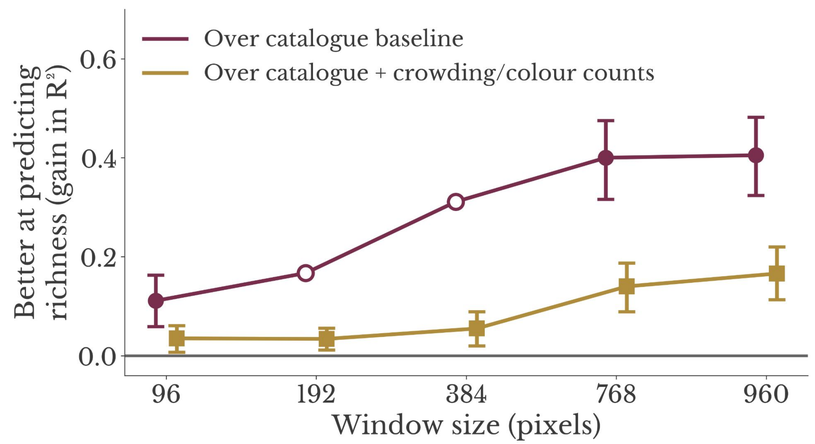
On centred objects, an adapter closed about half to two-thirds of the gap to an astronomy model (AION-1) on brightness and colour, but no model, including AION-1, reached our accuracy bars. So we asked a cheaper question: do wider windows help? We fed frozen DINOv3 windows of 96-960 px around each object to a simple model, and compared with baselines from catalogue properties and cheap image features.

## What we found

Cluster richness (how many galaxies a cluster holds; 1,962 DES Y3 clusters, exploratory): wider windows predict it far better, but simply counting red, crowded neighbours gets about 2/3 of that gain at 768 px.

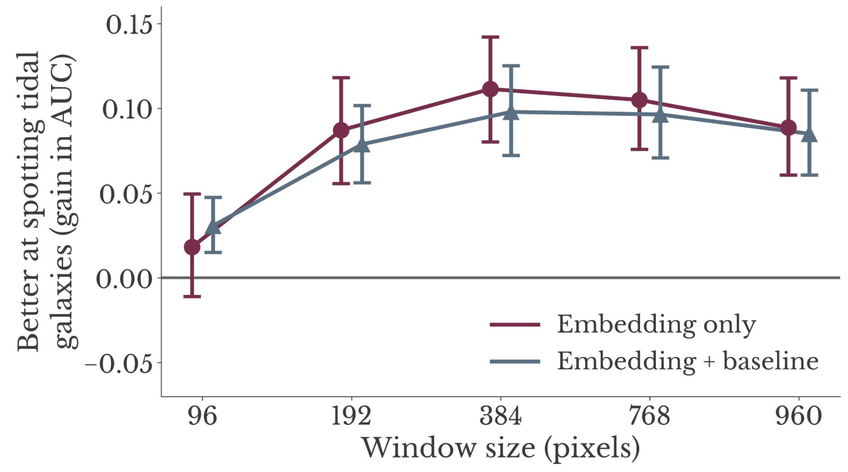
Tidal features (faint streams, tails and shells left by galaxy interactions; 2000 galaxies, 264 positives): from 192 px up, the model picks the tidal galaxy over a non-tidal one about 8-11 percentage points more often than the baseline. Bigger windows beyond that don't clearly help.

Details, data and code:
github.com/anaennis/build-or-adapt-poster



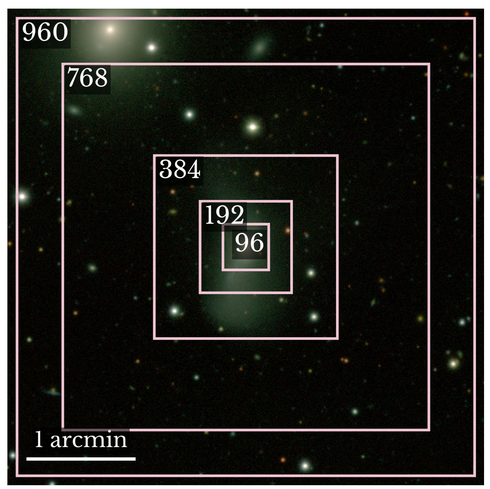

Exploratory, no pass bar. 1,962 redMaPPer clusters (DES Y3, z 0.25–0.35). Bars: logged intervals; hollow points have none logged.

Richness: wider windows help, but a cheap counter gets ~2/3 of the gain at 768 px



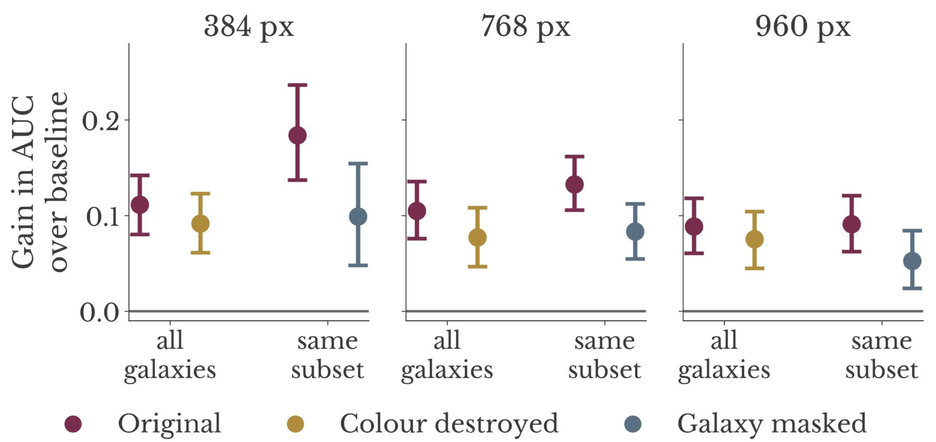

Pilot: 2000 galaxies, 264 positives, one run. Bars: 95% interval over sky regions. AUC = chance a tidal galaxy is ranked above a non-tidal one (0.5 = guessing).

Tidal features: better at spotting tidal galaxies from 192 px up (+0.08 to +0.11 AUC)




Windows tested, in pixels

Imaging: Legacy Surveys / D. Lang (Perimeter Institute), CC BY 4.0




The gain persists without colour, or with the galaxy masked